KFA – 2nd Semester 2025 – DCN – Mid-Term Practice Questions

1. **What is data?**
2. **What is data communication?**
3. **What is simplex transmission?**
4. **What is half-duplex transmission?**
5. **What is full-duplex transmission?**
6. **Describe the types of transmission media.**
7. **What is attenuation?**
8. **What is a network?**
9. **Explain PAN, SAN, LAN, MAN, and WAN with examples.**
10. **Differentiate between SAN and NAS.**
11. **Define circuit switching, packet switching, and message switching.**
12. **Differentiate between circuit switching, packet switching, and message switching.**
13. **Explain the advantages and disadvantages of circuit switching, packet switching, and message switching.**
14. **Describe datagram packet switching and virtual circuit packet switching.**
15. **Why is technology management important? Explain its objectives and key application areas.**
16. **Explain the pyramid of management and describe all its levels.**
17. **What are the roles of IT managers in networking?**
18. **What factors influence technology decisions? Explain.**
19. **Why are network policies, resource allocation, and planning necessary in an organization?**
20. **Explain the benefits of using a layered model in computer networks.**
21. **Describe all OSI layers and the services provided by each. Explain encapsulation and de-encapsulation.**
22. **Explain the benefits of segmentation of messages.**
23. **What is sequencing of a message? Explain with an example.**
24. **What is static IP addressing?**
25. **What is dynamic IP addressing?**
26. **Differentiate between static and dynamic IP addressing.**
27. **Describe all classes of IP addresses. Include IP ranges, starting bits, total networks, total hosts, and special addresses.**
28. **What are network ID, broadcast ID, and host ID? Explain with examples.**
29. **Why is a network gateway used?**
30. **What is subnetting? Why is it used? Explain its benefits.**
31. **Differentiate between classful and classless subnetting.**
32. **Compare routers with computers.**
33. **Describe router components and their functions.**
34. **Explain how a router initializes.**
35. **What are the functions of a router?**
36. **Which router models are suitable for campus, data center, and branch networks? Explain.**
37. **Explain how a router forwards traffic when a packet arrives.**
38. **What is meant by longest prefix match in routing? Explain with an example.**
39. **Calculate the network ID, broadcast ID, first usable IP address, last usable IP address, total IP addresses, and total usable host IP addresses for the following:**
    A. **192.168.5.5/29**
    B. **172.16.5.110/22**
    C. **172.17.10.138/17**
    D. **192.168.60.147/28**
40. **What are the benefits of using dynamic routing compared to static routing?**

41. **What is meant by gateway of last resort?**
42. **Describe the packet forwarding decision process in a router.**
43. **Explain all three packet forwarding mechanisms used by routers.**
44. **Write about the three routing table principles.**